

PRIYA RAMAN

Generative AI Architect & Engineer

📍 Aurora, IL | ✉️ priraman@gmail.com | ☎️ +1-920-664-2245 | 🔗 [linkedin.com/in/priya-raman](https://www.linkedin.com/in/priya-raman) 📁 [Technical Portfolio](#)

SUMMARY

GenAI Architect with 18+ years building enterprise-scale solutions in Healthcare, Retail & Supply Chain, and Banking. Transitioned from systems engineering to lead production-grade GenAI, expertise spans the full GenAI lifecycle: domain-specific LLM fine-tuning, distributed multi-agent orchestration, enterprise LLM platform engineering across GCP and Azure (Gemini & Anthropic Claude), and LLMOps with custom evaluation, prompt-cache optimization, and unified observability.

KEY COMPETENCIES

- **Enterprise LLM Platform Engineering & Cost Optimization (Azure):** Architecting production Anthropic Claude deployments on Azure AI Foundry, engineering prompt-cache optimization, two-pass multi-domain orchestration, and anti-hallucination evidence-sandboxing for regulated-industry document scoring; instrumenting Application Insights telemetry and pre-flight cost estimation for full session-cost visibility, with metadata-driven model routing and APIM-based inter-service authentication.
- **Production Multi-Agent Orchestration & Cloud Architecture:** Architecting high-reliability distributed systems using Google Agent Engine, ADK 1.20, and LangGraph. Proven ability to translate complex sequential/parallel LLM workflows into highly optimized, deployment-ready Vertex AI containers, utilizing Custom MCP Servers for low-latency external tool execution.
- **LLM Fine-Tuning & Lifecycle Optimization:** Orchestrating the full E2E journey from data augmentation and domain-specific PEFT (LoRA, Unsloth)—achieving up to 331% performance gains from LLM-as-a-Judge evaluation and scalable 3-tier GCP deployments with GGUF quantization.
- **Enterprise Observability:** Pioneering the "Telemetry Gravity" pattern to merge disparate instrumentation from LlamaIndex, GoogleGenAI, and ADK into a unified 'Single Pane of Glass' for debugging multi-agent workflows across local (Arize Phoenix) and GCP Cloud Trace.
- **High-Fidelity Reliability Engineering:** Engineering custom evaluation engines and trajectory validators to bypass ADK framework abstractions, ensuring 100% adherence to complex business requirements via Custom Manual Rubric Evaluators.
- **Advanced RAG & Retrieval Systems:** Constructing SOTA pipelines from scratch using hybrid search (Vector + BM25), LLM-powered query expansion, and Reciprocal Rank Fusion (RRF) for deeply grounded reasoning.
- **Scalable GCP LLMops & Deployment:** Implementing secure 3-tier GenAI apps using Vertex AI Endpoints, Cloud Run, API Gateway, managed via Docker, Artifact Registry, GCS, & gcloud CLI, with cloud-native troubleshooting.
- **Foundational ML & Data Analytics:** Executing in-depth EDA, data sourcing, and visualization (Matplotlib, Seaborn) to build high-performance classification, forecasting ML Models and CV models using transfer learning and automated hyperparameter tuning.

TECHNICAL STACK

- **Azure AI Platform & LLM Gateway:** Azure AI Foundry, Anthropic Claude (Opus, Sonnet), Azure API Management (APIM) AI Gateway, Application Insights, Azure Blob Storage, Anthropic Prompt Caching (ephemeral cache_control).
- **Model Routing, Governance & Inter-Service Auth:** Config-driven & APIM-policy metadata-based model routing, versioned prompt governance, APIM subscription-key/JWT-based inter-service authentication, AI/content-safety guardrail evaluation.
- **Agentic Orchestration & Protocols:** Google Gemini API, Vertex AI Agent Engine & Model Garden, Vertex AI Search + Grounding, Custom MCP Server(FastMCP), Remote API Tunneling(zrok/ngrok), Perplexity API, LangChain, LlamaIndex, Agent Development Kit (ADK 1.20), LangGraph, Hugging Face Agents, llama-cpp-python, Ollama, Mem0.
- **LLMops & Observability:** Telemetry Gravity Architecture, OpenTelemetry(OTEL) Instrumentation, Distributed Tracing GoogleGenAI, LlamaIndex, ADK), Arize Phoenix, LangFuse, GCP Cloud Trace & Logging, Weights & Biases
- **Reliability & Evaluation:** Custom Manual Rubric Evaluators, Trajectory Validation, "LLM-as-a-Judge" Frameworks, RRF (Reciprocal Rank Fusion) Hybrid Search, Golden Dataset Curation.
- **Model Engineering & LLM Fine-Tuning:** PEFT (LoRA, Unsloth), HuggingFace (Transformers, TRL), GGUF Quantization (llama-cpp), Keras Tuner (Neural Network Hypertuning), Dataset Augmentation.
- **Cloud Infrastructure & Deployment:** GCP (Cloud Run, Vertex AI Endpoints, API Gateway, Artifact Registry, GCS), Docker, FastAPI, Gradio, Streamlit, Next.js 15, gcloud CLI, Git
- **Foundational Systems & ML:** Python (AsyncIO, Pydantic), Swift (iOS Architecture), Oracle PL/SQL, SQL, Bash,

Scikit-learn, TensorFlow/Keras, XGBoost, CatBoost, LightGBM.

- **Data & Vector Architecture:** FAISS (Persistent Memory), ChromaDB, Pandas, NumPy, EDA, Hyperparameter Tuning

STRATEGIC GENAI CONSULTING (ENTERPRISE SOLUTIONS)

(CYCLOTRON | JUN 2026 – PRESENT)

Agentic SAD Review & Governance Framework

Automated Solution Architecture Document (SAD) scoring system for a Bank Client, evaluating design submissions at SDLC quality gates

Azure AI Foundry, Anthropic Claude, Azure API Management (APIM), Application Insights, Azure Blob Storage

- **Highlight: Prompt Cache Optimization:** Researched and implemented **industry-standard Anthropic prompt-caching** patterns (converged on independently, in parallel with published approaches such as **Uber Engineering's**); since the **full conversation history is resent on every turn**, implemented **caching between turns** so the **entire SAD document** is saved to cache once and **served from cache on every subsequent turn** rather than reprocessed; used **ephemeral cache breakpoints**, warm-cache priming calls, and **Turn-2 output capping** (`max_tokens=1`); measured **cache-hit rates up to 71.8%** (33-run mean **64.6%**) in production telemetry, materially cutting per-run token cost.
- **Telemetry & Session Cost Visibility:** Instrumented **Azure Application Insights** telemetry (run/job/correlation-ID tagged) feeding a **live HTML dashboard** (API calls, cache performance, domain breakdown, borrowed-evidence tracking), alongside a zero-API-call **pre-flight cost estimator** accurate to within **approximately 2%** of the actual invoice, giving full session-level cost visibility.
- **Two-Pass Orchestration & Model Routing:** Architected a **two-pass pipeline**, heavier-model (**Opus**) evidence extraction in Pass 1 delegating to **7 parallel per-domain scoring calls** on an operation-efficient model (**Sonnet**) in Pass 2 for latency; implemented **dual-layer, SAD-metadata-driven model routing** (an **APIM-policy** proof of concept plus the authoritative config-driven routing in code) and **inter-service authentication** (APIM subscription-key/JWT-based access control) so only APIM is authorized to call **Azure AI Foundry**.
- **Anti-Hallucination & Evidence Traceability:** Enforced **strict JSON-only, evidence-only outputs** in Pass 1 and sandboxed each Pass 2 domain to its own evidence plus a deterministic **cross-domain evidence-borrowing backfill**, preventing mishandled or corrupted sections from contaminating unrelated domain scores; ensured **full evidence traceability** by linking every rubric criterion's score back to its supporting source evidence.
- **Config-Driven Governance:** Split the EA-owned scoring rubric from engineering configuration to eliminate **17+ hardcoded values**, and established **versioned, immutable prompt governance** with audit-trail stamping; partnered on AI/content-safety guardrail evaluation across **four candidate enforcement layers** (app/agent, Foundry-model, APIM gateway, model provider), confirming **APIM gateway enforcement** as the right layer for this architecture.

(KSAP TECHNOLOGIES | FEB 2024 – MAR 2026)

Google ADK, Agent Engine, LlamaIndex, OpenTelemetry(OTEL), GCP, Vertex AI, Cloud Run, API Gateway

- **Highlight: Autonomous Ingestion & Cloud Migration :** Engineered production-grade autonomous agents and high-fidelity workflows to automate the extraction of complex domain-specific documents and legacy-to-cloud data migrations with 100% logic adherence.
- **Self-Healing Agents & Unified Observability:** Deployed unified, fault-tolerant observability and self-monitoring QA agents to secure enterprise-wide deployment-readiness.

APPLIED GENAI ENGINEERING (OPEN-SOURCE IMPLEMENTATIONS)

(CORE ENGINEERING PORTFOLIO | MAR 2025 – PRESENT)

Distributed Multi-Agent Research Orchestrator

A multi-agent research assistant inspired by Google's NotebookLM

ADK, LangGraph , LlamaIndex , MCP, MLX Model, GCP(Vertex AI, AgentEngine), Arize Phoenix, Docker  [GitHub](#)

- **Highlight: Hybrid Orchestration & Multi-Agent Security:** Achieved **<2 min generation for 15 Min podcasts** via remote MCP/MLX; **Bridged** sync UI threads with async LLM endpoints to eliminate thread starvation; **Secured** production toggles(Gemini/Vertex Model Garden) by decoupling control/data plane identities via Secret Manager.
- **Highlight: High-Fidelity Reliability & Evaluation -** Achieved an **aggregate 91% pass rate** by architecting a **Custom Evaluation Framework** using Manual Rubric Criterion; **resolved core SDK limitations** by injecting JSON-based business logic and custom capture layers to monitor intermediate tool trajectories.
- **Enterprise Unified Telemetry:** Pioneered the **"Telemetry Gravity"** pattern using OTEL to merge disparate instru-

mentation from GoogleGenAI, ADK, and LlamaIndex into a "Single Pane of Glass" - GCP Cloud Trace/Arize Phoenix.

- **Serverless Production Infrastructure Optimization** : Engineered a multi-layer serverless topology on GCP (Cloud Run, Vertex AI); Eliminated **"Cold Start" and memory failures** by shrinking ephemeral containers to **<8MB by intense payload dieting**; Enforced **deterministic dependency resolution** to hydrate isolated containers within the Google build-farm's strict startup timeouts
- **Advanced Agentic RAG & Multi-Format Delivery**: Engineered a **SOTA retrieval engine - Reciprocal Rank Fusion (RRF) & hierarchical chunking** to provide agents with fault-tolerant context pipelines, seamlessly orchestrating PDFs into conversational chat, structured study guides, audio podcasts, and distinct visual LangGraph mindmaps.

Specialized Drug Interaction LLM: End-to-End Fine-Tuning & GCP Deployment

An Interactive AI Chatbot Showcasing MLOps from Training to Production on Google Cloud

Llama3.1 8B, Unsloth, PEFT(LoRA), SFT, Hugging Face(TRL), GGUF Quantization, GCP & Vertex API [GitHub](#)

- **Highlight: +331% performance gain** by fine-tuning of Llama-3.1-8B; **Augmented** niche DDI PyTDC datasets for drug-interaction precision; **Validated** outcomes using custom **LLM-as-a-Judge** evaluation
- **Highlight: Scalable Cloud Architecture**: Deployed a secure **3-tier MLOps** environment: **Vertex AI Endpoints (T4 GPUs)** for inference, **Cloud Run** for serverless UI (scale-to-zero), **API Gateway** for IAM/ID Token-protected access.
- **Hardware-Optimized Fine-Tuning**: **Maximized** VRAM/disk utilization on a single **RTX 4090** via **Unsloth and PEFT/LoRA**; Leveraged **Quantized base models** for high-speed, resource-efficient training; **Monitored** hardware telemetry to eliminate bottlenecks in high-stakes training environments.
- **Advanced Inference**: **Quantized** fine-tuned models into multiple **GGUF formats** for optimized inference; Managed **containerization (Docker)**, artifact storage (**GCS/Artifact Registry**), automated deployment (**gcloud CLI**).
- **Cloud Infrastructure Engineering**: Resolved **Vertex AI cold starts** and **504 timeouts** by optimizing model download latency; debugged **403 inter-service authentication failures** and container payload mismatches; navigated GPU quota limits and GCS permissions.

Generative Bid Agent: Enterprise RFP Intelligence System

Google ADK, Vertex AI Search, Gemini Pro, Next.js 15, Docker

[GitHub](#)

- **Highlight: Multi-Modal RAG with "Triangle of Truth" Reasoning**: Engineered Cross-Document RAG agent using **Vertex AI Search** to synthesize heterogeneous documents (PDF/Excel/PPT), deriving estimation heuristics and generating solution architecture decks—automating end-to-end RFP response generation
- **Semantic Retrieval with Dynamic Filtering & Re-Ranking**: Implemented a 2-stage retrieval pipeline with dynamic metadata filters(Client ID, domain) and **Vertex AI Ranking API for contextual re-ranking across heterogeneous enterprise documents**.

Priya's AI Portfolio & Reasoning Engine (The Digital Twin)

Next.js 15, Gemini 2.5 Flash, Agentic Context Injection, Strict System Guardrails, Cloud Run, Docker [GitHub](#)

- **Highlight: Agentic Digital Twin & AI Guardrails**: Synthesized 18+ years of professional history into **structured JSON memory for Gemini-2.5-Flash**; Engineered **strict refusal heuristics and prompt boundaries** to guarantee 100% professional integrity.
- **Cloud Architecture & FinOps**: Deployed to **Cloud Run** with secure API key injection(**Secret Manager**) and optimized **infrastructure costs** via **scale-to-zero** FinOps logic.

GAIA Agent: General AI Assistant Orchestrator

LangGraph, Mem0, Gemini API

[GitHub](#)

- **Highlight: SOTA Benchmark Performance**: Achieved **100% accuracy on the GAIA (General AI Assistant) benchmark** by engineering a sophisticated agentic architecture capable of multi-step tool-use and reasoning
- **Memory & Multi-Tool Integration**: Leveraged **Mem0** for long-term persistent memory and a suite of **12 heterogeneous tools (Tavily, Wikipedia, Multimedia/PDF Processors)** managed via a **LangGraph StateGraph** to solve high-complexity information retrieval tasks.

FOUNDATIONAL AI/ML PROJECTS (COURSE TENURE | JUL 2024 – APR 2025)

LLM Powered Financial Market: Sentiment Analysis & News Summarization

[GitHub](#)

- **Highlight: Achieved 49.2% F1-score** on imbalanced data by evaluating **Sentence-Transformer, Word2Vec, and GloVe** embeddings across 6+ ML models (**LightGBM, CatBoost**); Engineered an automated news summarizer utilizing **quantized Mistral-7B** for actionable investment insights.

Plant Seedlings Image Classification using Convolutional Neural Networks (CNNs)

GitHub

- **Highlight:** Increased accuracy 83.1% → 91.1% by VGG19 transfer learning using TensorFlow & Keras; optimized via Grid Search (Optimizer/LR), Data Augmentation, and Batch Normalization to resolve high-variance generalization issues in noisy plant-seedling image datasets.

Deep Learning for Customer Churn Forecasting in Retail Banking

GitHub

- **Highlight:** Drove 86.8% recall using the Keras Tuner framework for automated hyperparameter searches; identified optimal SGD with momentum, learning rate, & dropout configurations for at-risk customer identification.

Credit Card Churn Prediction with Supervised Classification

GitHub

- **Highlight:** Achieved 96.8% recall(85.9% F1) using Gradient Boosting with an AdaBoost base estimator; engineered a suite of ensemble models (XGBoost, Bagging) with undersampling to counteract severe class imbalance.

Loan Propensity Modeling with Predictive Classification

GitHub

- **Highlight:** Reached 96.5% recall (85% F1) via Decision Tree pre-pruning & custom class weights; pushed boundary metrics for targeted marketing and increased loan conversion using model-derived customer segments.

RELEVANT PROFESSIONAL EXPERIENCE

Mobile Principal Engineer I	CVS Health	[OCT 2021 – OCT 2023]
Mobile Architect/Lead	Capgemini - McDonalds	[JAN 2019 – OCT 2021]
Mobile Developer	Keevat LLC & Accenture - Allstate	[APR 2015 – DEC 2018]
• Systems Architecture & Scale: Engineered high-availability platforms for millions of users; Applied distributed patterns and concurrency now utilized in production-grade 3-tier GenAI and LLMOps pipelines.		
Facets ERP Oracle Developer	Cognizant Technology Solutions	[SEP 2006 – DEC 2012] & [APR 2014 – APR 2015]
• Data Engineering Foundation: Managed large-scale DBs and complex data pipelines; Pioneered data lineage & wrangling workflows required for modern RAG and vector databases optimizations.		

EDUCATION

PGP, Artificial Intelligence & Machine Learning	UT Austin, McCombs	[July 2024 – April 2025]
Bachelor of Engineering, Electronics & Communication	ACCET, Anna University	[2002 – 2006]

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

- **Google Cloud: Multi-Agent Architectures** (ADK & Agent Engine); **AgentVerse Workshop** (Architected production-ready autonomous agents via MLOps and CI/CD containerization) [2025]
- **LLM & Agent Mastery:** Hugging Face (Agents | LLM Fundamentals | Fine-Tuning LLMs); Kaggle (Google 5-Day GenAI Intensive | Capstone Portfolio); ExamPro (GenAI Bootcamp | Badge) [2025]
- **Claude Certified Associate – Foundations** – Anthropic [Aug 2026]